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IMAGE SENSOR WITH PHASE DETECTION AUTOFOCUS PIXEL

Abstract

A phase detection autofocus (PDAF) pixel cell and an image sensor including a PDAF pixel cell is described. The PDAF pixel cell includes a photosensitive region including one or more photodiodes formed within a semiconductor material proximate to a backside surface of the semiconductor material, a microlens optically aligned with the photosensitive region; and a spectral filter disposed between the microlens and the metal shield. A first lateral width of the spectral filter is less than a second lateral width of the metal shield along a direction parallel to the backside surface to form a gap extending from an edge of the spectral filter toward an edge of the metal shield when viewed from a plan view.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Application No. 63/560,375, filed Mar. 1, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD
[0002] This disclosure relates generally to image sensors, and in particular but not exclusively, relates to phase detection autofocus (PDAF) pixels for image sensors.
BACKGROUND INFORMATION
[0003] Image sensors are one type of semiconductor device that have become ubiquitous and are now widely used in digital cameras, cellular phones, security cameras, as well as, medical, automobile, and other applications. As image sensors are integrated into a broader range of electronic devices it is desirable to enhance their functionality, performance metrics, and the like in as many ways as possible (e.g., resolution, power consumption, dynamic range, size, etc.) through both device architecture design as well as image acquisition processing. However, it is appreciated that many of these metrics are inversely related. For example, pixel size may be increased to improve dynamic range but have increased noise. In another example, resolution may be increased by increasing the number of pixels, but if pixel size is maintained then the physical size of the image sensor increases. Accordingly, improving one or more performance metrics of semiconductor devices such as image sensors while mitigating adverse effects on other performance metrics remains challenging.
[0004] The typical image sensor operates in response to image light reflected from an external scene being incident upon the image sensor. The image sensor includes an array of pixels having photosensitive elements (e.g., photodiodes) that absorb a portion of the incident image light and generate image charge upon absorption of the image light. The image charge photogenerated by the pixels may be measured as analog output image signals on column bitlines that vary as a function of the incident image light. In other words, the amount of image charge generated is proportional to the intensity of the image light, which is read out as analog image signals from the column bitlines and converted to digital values to produce digital images (i.e., image data) representative of the external scene.
[0005] In some imaging applications, image sensors may employ a technique known as phase detection autofocus (PDAF) to perform autofocus by measuring an offset or phase difference between two image signals of the same object or external scene generated from different perspectives. The two image signals may be used to determine a relative position between an object of interest and a focal plane of the image sensor. The computed magnitude and sign of the calculated phase difference may be correlated to an amount of defocus that can be used to estimate an amount of lens displacement needed to have the image in focus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0006] Non-limiting and non-exhaustive embodiments of the invention are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified. Not all instances of an element are necessarily labeled so as not to clutter the drawings where appropriate. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles being described.

[0007] FIG. **1** illustrates an example pixel array that includes a plurality of quad phase detection (QPD) pixel cells, in accordance with teachings of the disclosure.
[0008] FIG. **2**A illustrates an example layout showing a partial pixel cell array for an image sensor featuring a dedicated phase detection autofocus (PDAF) pixel cell, in accordance with teachings of the disclosure.
[0009] FIG. **2**B illustrates a cross-sectional view of the partial pixel cell array illustrated in FIG. **2**A along cutline A-A′, in accordance with teachings of the disclosure.
[0010] FIG. **3** illustrates a cross-sectional view and a plan view of a partial pixel cell array of an image sensor including a PDAF pixel cell with reduced induced crosstalk, in accordance with teachings of the disclosure.
[0011] FIG. **4**A illustrates an exemplary partition of a pixel cell array that includes a plurality of PDAF pixel cells into regions of interests (ROI), in accordance with teachings of the disclosure.
[0012] FIG. **4**B illustrates a first pair of PDAF pixel cells including a left-oriented PDAF pixel cell and a right-oriented PDAF pixel cell located within any of ROI **1**, ROI **2**, or ROI **3** illustrated in FIG. **4**A, in accordance with teachings of the disclosure.
[0013] FIG. **4**C illustrates a second pair of PDAF pixel cells including a left-oriented PDAF pixel cell and a right-oriented PDAF pixel cell located within any of ROI **4**, ROI **5**, or ROI **6** illustrated in FIG. **4**A, in accordance with teachings of the disclosure.
[0014] FIG. **4**D illustrates a third pair of PDAF pixel cells including a left-oriented PDAF pixel cell and a right-oriented PDAF pixel cell located within any of ROI **7**, ROI **8**, or ROI **9** illustrated in FIG. **4**A, in accordance with teachings of the disclosure.
[0015] FIG. **4**E illustrates a more detailed view of left and right oriented PDAF pixel cells illustrated in FIG. **4**B-**4**E, in accordance with teachings of the disclosure.
[0016] FIG. **5** illustrates a cross-sectional view and a plan view of a partial pixel cell array of an image sensor including a PDAF pixel cell with reduced induced crosstalk, in accordance with teachings of the disclosure.
[0017] FIG. **6** illustrates left and right oriented PDAF pixel cells with two spectral filters disposed over corresponding metal shields, in accordance with teachings of the disclosure.
[0018] FIG. **7** illustrates a cross-sectional view and a plan view of a partial pixel cell array of an image sensor including a PDAF pixel cell with a combination of a spectral filter and an antireflection layer for reduced induced crosstalk, in accordance with teachings of the disclosure.
[0019] FIG. **8** illustrates a block diagram of an imaging system for an image sensor, in accordance with teachings of the disclosure.

DETAILED DESCRIPTION

[0020] Embodiments of an apparatus, system, and method each related to an image sensor with a phase detection autofocus (PDAF) pixel are described herein. In the following description, numerous specific details are set forth to provide a thorough understanding of the embodiments. One skilled in the relevant art will recognize, however, that the techniques described herein can be practiced without one or more of the specific details, or with other methods, components, materials, etc. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring certain aspects.
[0021] Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.
[0022] It will be understood that, although the terms first, second, third, etc., may be used in the disclosure and claims to describe various elements, these elements should not be limited by these terms and should not be used to determine the process sequence or formation order of associated

elements. Unless indicated otherwise, these terms are merely used to distinguish one element from another element. Thus, a first element discussed below could be termed a second element without departing from the teachings of the disclosed embodiments.

[0023] Throughout this specification, several terms of art are used. These terms are to take on their ordinary meaning in the art from which they come, unless specifically defined herein or the context of their use would clearly suggest otherwise. It should be noted that element names and symbols may be used interchangeably through this document (e.g., Si vs. silicon); however, both have identical meaning.

[0024] Spatially relative terms, such as "beneath", "below", "lower", "under", "above", "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as "below" or "beneath" or "under" other elements or features would then be oriented "above" the other elements or features. Thus, the exemplary terms "below" and "under" can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. In addition, it will also be understood that when a layer is referred to as being "between" two layers, it can be the only layer between the two layers, or one or more intervening layers may also be present.

[0025] Further, it will be understood that when an element is referred to as being "connected," or "coupled," to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being "directly connected," or "directly coupled," to another element, there are no intervening elements present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., "between," versus "directly between," "adjacent," versus "directly adjacent," etc.).

[0026] Further still, it will be understood that when an element or layer is referred to as being "formed on," another element or layer, it can be directly or indirectly formed on the other element or layer. That is, for example, intervening elements or layers may be present. In contrast, when an element or layer is referred to as being "directly formed on," to another element, there are no intervening elements or layers present. Other words used to describe the relationship between elements or layers should be interpreted in a like fashion (e.g., "between," versus "directly between," "adjacent," versus "directly adjacent," etc.).

[0027] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of example embodiments of the invention. As used herein, the singular forms "a," "an," and "the," are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises," "comprising," "includes," and/or "including," when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0028] The term "have", "may have", "include", "may include" or "comprise" used herein indicates the existence of a corresponding feature (e.g., a number, a function, an operation, or an element) and does not exclude the existence of an additional feature.

[0029] The term "A or B", "at least one of A and/or B", or "one or more of A and/or B" may include all possible combinations of items listed together. For example, the term "A or B", "at least one of A and B", or "at least one of A or B" may indicate all the cases of (1) including at least one A, (2) including at least one B, and (3) including at least one A and at least one B.

[0030] It will be understood that when a certain element (e.g., a first element) is referred to as being "operatively or communicatively coupled with/to" or "connected to" another element (e.g., a second element), the certain element may be coupled to the other element directly or via another element (e.g., a third element). However, when a certain element (e.g., a first element) is referred to as being "directly coupled" or "directly connected" to another element (e.g., a second element), there may be no intervening element (e.g., a third element) between the element and the other element.

[0031] The term "configured (or set) to" may be interchangeably used with the term, for example, "suitable for", "having the capacity to", "designed to", "adapted to", "made to", or "capable of". The term "configured (or set) to" may not necessarily have the meaning of "specifically designed to". In some cases, the term "device configured to" may indicate that the device "may perform" together with other devices or components. For example, the term "processor configured (or set) to perform A, B, and C" may represent a dedicated processor (e.g., an embedded processor) for performing a corresponding operation, or a generic-purpose processor (e.g., a CPU or an application processor) for executing at least one item of software or program stored in a memory device to perform a corresponding operation.

[0032] The term "vertical" refers to a direction that is perpendicular to a surface of a substrate or material layer.

[0033] The term "high-K material" herein refers to a material having a dielectric constant that is greater than 3.9. In some embodiments, term "high-K material" may also refer to a material having a dielectric constant that is greater than a dielectric constant of the silicon dioxide (SiO2).

[0034] The term "p-type" defines a structure, layer, and/or region in a substrate material layer (e.g., semiconductor substrate or epitaxial layer) as being doped with p-type dopant, such as boron.

[0035] The term "n-type" defines a structure, layer, and/or region in a substrate material layer (e.g., semiconductor substrate or epitaxial layer) as being doped with n-type dopant, such as phosphorus and/or arsenic.

[0036] In some embodiments, the term "about" and the term "substantially" can refer to a value of a given quantity or manufacturing parameters that varies within 5% of the value (e.g., ±1%, ±2%, ±3%, ±4%, ±5% of the value). These values are merely examples and are not intended to be limiting. The terms "about" and "substantially" can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings of the disclosure.

[0037] FIG. **1** illustrates an example pixel array **100** that includes a plurality of quad phase detection (QPD) pixel cells **110**, in accordance with teachings of the disclosure. As illustrated in FIG. **1**, QPD pixel cells **110** are arranged in accordance to a Bayer pattern with each QPD pixel cell including four pixels (e.g., photodiodes) that share a same color filter (e.g., "R" for red color filters, "G" for green filters, and "B" for blue color filters) and further share a same microlens included in plurality of microlenses **120**. Advantageously, plurality of QPD pixel cells **110** may be used for both imaging and autofocus when each individual pixel included in plurality of QPD pixel cells **110** are being read out. Specifically, a single microlens optically aligned with a group of photodiodes as illustrated in FIG. **1** enables images signals of different perspectives to be captured (e.g., to employ phase difference autofocus techniques by comparing, for example, image signals generated with left pixels to image signals generated with right pixels).

[0038] An image sensor with plurality of QPD pixel cells **110** may be operated in a high resolution mode (e.g., a "1C" mode) or a binning mode (e.g. a "4C" mode). When operating in the high resolution mode, each individual photodiode or pixel included in plurality QPD pixel cells **110** is readout to achieve high resolution. When operating in the binning mode, multiple photodiodes of a given pixel cell included in plurality of QPD pixel cells **110** are binned together for readout. For example, during the 4C mode (e.g., 2-by-2 binning mode), four photodiodes (e.g., four photodiodes of a given pixel cell included in plurality of QPD pixel cells **110**) of a same color are binned together for readout. Modes of operation may be selected based on target pixel performance (e.g., 1C mode offers lower noise performance relative to 4C mode while 4C mode offers faster readout relative to 1C mode).

[0039] However, autofocus remains limited when operating plurality of QPD pixel cells **110** in 4C mode since, for example, there is a lack of an aperture to split light for generating phase difference images (e.g., each photodiode included in a given pixel cell is binned for readout meaning plurality of QPD pixel cells **110** may not be able to form left and right images for phase detection). In other words, phase detection autofocus may not be readily employed using plurality of QPD pixel cells **110** operating in 4C mode. Embodiments of the disclosure included dedicated phase detection pixel cells compatible with, but not necessarily exclusively for, image sensors operating in a binning mode (e.g., 4C mode).

[0040] FIG. **2**A illustrates an example layout showing a partial pixel cell array for an image sensor featuring a dedicated phase detection autofocus (PDAF) pixel cell, in accordance with teachings of the disclosure. As illustrated, pixel cell array **200** includes a plurality of QPD pixel cells **210** arranged in rows and columns in a Bayer pattern. Plurality of QPD pixel cells **210** generates electrical or image signals in response to incident light (e.g., light **270** illustrated in FIG. **2**B). Individual pixel cells included in plurality of QPD pixel cells **210** each include four photodiodes (i.e., PD**1**-PD**4**) of a same color (e.g., “R”, “G”, or “B”, which respectively correspond to red, green, or blue pixels) under a single or same microlens **214**. It is appreciated that in some embodiments plurality of QPD pixels **210** may be arranged based on non-Bayer pattern such as red, green, blue and infrared, or red, green, blue, and clear, etc.). Microlens **214** may be configured with an appropriate size to cover or otherwise be optically aligned with the photosensitive region of a respective pixel cell (e.g., an individual microlens of a given pixel cell is optically aligned with or otherwise disposed over all photodiodes included in the given pixel cell for each pixel cell included in plurality of QPD pixel cells **210**). In some embodiments, individual pixel cells of plurality of QPD pixel cells **210** are referred to as imaging pixel cells. However, it is appreciated that the imaging pixel cells may be also used for phase detection autofocus when operating in high resolution mode.

[0041] To enable autofocus when operating in binning mode (e.g., 4C mode), pixel cell array **200** includes sparsely distributed phase detection autofocus pixel cells **220** dedicated to phase detection. As illustrated, PDAF pixel cells **220** corresponds to QPD pixel cells with a clear or omitted color filter having half of the pixels (e.g., a subgroup of two photodiodes on the left or right side) shielded (e.g., via metal or opaque shield **220**) to create an asymmetry to generate left or right images (e.g., left and right oriented metal shields may be used to form left and right oriented PDAF pixel cells **220** that may be respectively used to generate left and right image signals that may be compared for phase detection).

[0042] However, it has been found that PDAF pixel cells **220** may lead to crosstalk affecting sensitivity of neighboring normal color imaging pixels (i.e., plurality of QPD pixel cells **210** including first level pixel **230**, second level pixel **250**, third level pixel **260**, and/or corner pixel **240**). It has been observed that metal shield **220** used to block light for PDAF pixel cells **220** result in notable levels of optical crosstalk to proximate pixel cells included in plurality of QPD pixel cells **210** due to reflection and/or refraction. Such crosstalk may be referred to as phase detection (PD) crosstalk. PD crosstalk herein may be defined as the sensitivity impact that a metal shield based phase detection pixel (i.e., PDAF pixel cells **220**) has on surrounding or nearby imaging pixel cells (e.g., first level pixel **230**, second level pixel **250**, third level pixel **260**, and/or corner pixel **240**).

[0043] FIG. **2**B illustrates a cross-sectional view of the partial pixel cell array **200** illustrated in FIG. **2**A along cutline A-A′, in accordance with teachings of the disclosure. PDAF pixel cell **220** includes four photodiodes (e.g., a first photodiode PD**1**, a second photodiode PD**2**, a third photodiode PD**3**, and a fourth photodiode PD**4**) arranged in a two-by-two array to form left photodiodes PDL and right photodiodes PDR formed in semiconductor material **202**. A microlens **212** is optically aligned with left photodiodes PDL and right photodiodes PDR to direct incident light towards left photodiodes PDL and right photodiodes PDR. PDAF pixel cell **220** may be referred to as a half-shield phase detection autofocus pixel cell since metal shield **225** (e.g., formed of a metal material to create an opaque layer that includes a metal material such as aluminum or tungsten) covers half of the photodiodes (e.g., left photodiodes corresponding to PDL) included in PDAF pixel cell **220** while leaving the remaining photodiodes of PDAF pixel cell **220** uncovered (e.g., right photodiodes corresponding to PDR). In the illustrated embodiment, metal shield **225** is disposed between the microlens **212** and left photodiodes PDL to block incident light from entering the left half of a photosensitive region (e.g., corresponding to left photodiodes PDL and right photodiodes PDF, collectively) of PDAF pixel cell **220**. It is appreciated that the orientation of metal shield **225** may be flipped (e.g., to cover right photodiodes PDR while leaving left photodiodes PDL uncovered) such that left and right image signals may be generated and compared for phase detection autofocus. In other words, left-shielded PDAF pixel cells and right-shielded PDAF pixel cells may be used in combination for phase detection autofocus to determine how much displacement of a lens is necessary to focus on a particular point of interest.

[0044] However, it has been found that metal shield **225** for PDAF pixel cell **220** may cause optical crosstalk to surrounding imaging pixels (e.g., plurality of QPD pixel cells **210**) that are positioned in proximity to PDAF pixel cell **220** (e.g., first level pixels **230**, second level pixels **250**, third level pixels **260**, and/or corner pixels **240**) due, for example, to reflection and/or refraction associated with PDAF pixel cell **220**. Such crosstalk may be referred to as phase detection (PD) crosstalk. PD crosstalk herein may be defined as the sensitivity impact that PDAF pixel cells **220** has on surrounding or nearby imaging pixels included in plurality of QPD pixel cells **220** (e.g., relative to pixels included in plurality of QPD pixel cells **220** that are farther away and unaffected by PDAF pixel cells **220**). For example, pixel cell **229**B (e.g., including photodiode PD**1** **262**), pixel cell **231**G (e.g., including photodiodes PD**1** **232** and PD**2** **252**), pixel cell **233**G (e.g., including photodiodes PD**1** **254** and PD**2** **234**), and pixel cell **235**B (e.g., including photodiode PD**2** **264**) may each have elevated crosstalk levels due, at least in part, to PDAF pixel cell **220**.

[0045] In one example, incident light ray **270** may reflect off metal shield **225** to form reflected light ray **270**R**1** that follows a curvature of microlens **214** associated with an adjacent pixel cell included in plurality of QPD pixel cells **210** (e.g., pixel cell **231**G) and may result in multiple reflected light rays (e.g., rays **270**R**2** and **270**R**3**) directed towards the photosensitive regions of nearby pixels (e.g., photodiode PD**2** **252** and/or photodiode PD**1** **262**) In other words, light incident on metal shield **225** included in PDAF pixel cell **220** may be reflected, scattered, diffracted and subsequently sensed by photodiodes included in neighboring pixel cells included in QPD pixel cells **210**. For example, it has been found that first level pixels **230** that are immediately adjacent to PDAF pixel cell **220** (e.g., directly adjacent to PDAF pixel cell **220** along a same column or same row) may have at least a 5% signal crosstalk level, second level pixels **250** (e.g., pixels directly adjacent to first level pixels **230**) and/or corner pixels **240** (e.g., diagonally adjacent to PDAF pixel cell **220**) may have at least a 2% signal crosstalk level, and/or third pixel cells **260** (e.g., directly adjacent to second level pixels **250**) may have at least a 1% signal crosstalk level associated with PDAF pixel cell **220**. It is appreciated that the aforementioned signal crosstalk levels associated with PD crosstalk correspond to the amount of deviation in crosstalk relative to other QPD pixel cells included in plurality of QPD pixel cells **210** that are not influenced by PDAF pixel cell **220**. Embodiments of the disclosure provide an improved PDAF pixel cell relative to PDAF pixel cell **220** that mitigates the issue of PD crosstalk while also being compatible with binning modes of operation.

[0046] FIG. **3** illustrates a cross-sectional view **300**-X and a plan view **300**-P of a partial pixel cell array of an image sensor including a PDAF pixel cell **320** with reduced induced crosstalk, in accordance with teachings of the disclosure. PDAF pixel cell **320** represents an improved phase detection structure relative to PDAF pixel cell **220** illustrated in FIG. **2**A-**2**B. In the illustrated embodiment, PDAF pixel cell **320** includes a plurality of photodiodes, including left photodiodes PDL and right photodiodes PDR (e.g., first, second, third, and forth photodiodes arranged in a two-by-two array) surrounded by imaging pixels (e.g., pixel cell **329**B including photodiode PD**1** **362**, pixel cell **331**G including photodiode PD**2** **352** and PD**1** **332**, pixel cell **333**G including photodiode PD**2** **334** and PD**1** **354**, and pixel cell **335**B including photodiode **364**) formed within semiconductor material **302**. PDAF pixel cell **320** further includes metal shield **325**, spectral filter **340**, and microlens **350-1**.

[0047] PDAF pixel cell **320** includes a photosensitive region **309** (e.g., defined by isolation structure **304**) formed in semiconductor material **302** corresponding to regions where left photodiodes PDL and right photodiodes PDR are located (e.g., doped regions of semiconductor material **302** that collectively form left photodiodes PDL and right photodiodes PDR). In other words, PDAF pixel cell **320** includes photosensitive region **309** including one or more photodiodes (e.g., left photodiodes PDL and right photodiodes PDR) formed within semiconductor material **302** proximate to a backside surface BS of semiconductor material **302**. Semiconductor material **302** includes front side surface FS and the backside surface BS opposite of the front side FS. It is appreciated, backside surface BS may sometimes be referred to as an illuminated side and front side surface FS may be referred to as a non-illuminated side. As illustrated, microlens **350-1** is optically aligned with photosensitive region **309** (e.g., to direct light towards left photodiodes PDL and right photodiodes PDR). Metal shield **325** is disposed between microlens **350-1** and photosensitive region **309** (e.g., to block incident light from reaching left photodiodes PDL or right photodiodes PDR depending on an orientation of PDAF pixel cell **320**).

[0048] In illustrated embodiment, PDAF pixel cell **320** may be laterally surrounded by a plurality of QPD pixel cells (e.g., pixel cells **329**B, **331**G, **333**G, **335**B) and may be dedicated for phase detection. In other words, PDAF pixel cell **320** may provide phase detection for the plurality of QPD pixel cells operating in binning mode. Photodiodes included in plurality of QPD pixel cells and PDAF pixel cell **320** correspond to doped regions disposed or formed within semiconductor material **302**. In some embodiments, semiconductor material **302** includes or is otherwise formed of silicon, a silicon germanium alloy, germanium, a silicon carbide alloy, an indium gallium arsenide alloy, any other alloys formed of III-V group compounds, combinations thereof, one or more epitaxial layers of the aforementioned materials, or a bulk substrate thereof. More specifically, semiconductor material **302** may correspond to any semiconductor material or combination of materials that may be doped or otherwise configured to facilitate the formation of an integrated circuit (e.g., individual circuitry components such as source/drain regions of transistors, memory elements, photodiodes, or the like). In one embodiment, semiconductor material **302** corresponds to an epitaxial layer (e.g., P-type silicon layer or N-type silicon layer). In such an embodiment, photodiodes (e.g., left photodiodes PDL, right photodiode PDR, or other photodiodes annotated through the disclosure as PD**1**, PD**2**, PD**3**, or PD**4**) throughout the disclosure may be formed in the epitaxial layer corresponding to semiconductor material **302**.

[0049] It is appreciated that the term “photodiode” (e.g., left photodiodes PDL, right photodiode PDR, or other photodiodes annotated through the disclosure as PD**1**, PD**2**, PD**3**, or PD**4**) correspond to a doped region (e.g., formed via implantation) disposed within or otherwise surrounded by an oppositely doped region to form a photosensitive area capable of photogenerating image charge in response to incident light. For example, individual photodiodes included in left photodiodes PDL and/or right photodiodes PDR may correspond to an N-type semiconductor region (e.g., N-doped silicon region) disposed within a P-type semiconductor material (e.g., P-type doped silicon corresponding to semiconductor material **302**). Accordingly, in some embodiments individual photodiodes included in PDAF pixel cell **320** illustrated in FIG. **3** or other photodiodes throughout the disclosure each include a doped region that is oppositely doped (e.g., opposite conductivity type) relative to a doping type of semiconductor material **302** to form a PN junction for photo-sensing operation.

[0050] In the illustrated embodiment, pixel cell array **300** includes isolation structure **304** (e.g., a plurality of deep trench isolation structures) formed in semiconductors material **302** to provide isolation between individual photodiodes (e.g., left photodiodes PDL and right photodiodes PDR may be isolation from each other via isolation structure **304**) included in phase detection autofocus pixel cells and normal imaging pixel cells (e.g., pixel cells **329**B, **331**G, **333**G, **335**G, and so on). In some embodiments, isolation structure **304** may be arranged to form a grid defining photosensitive areas of individual photodiodes and/or pixel cells. As illustrated, isolation structure extends from a backside surface BS of semiconductor material **302** toward front side surface FS of semiconductor material **302**. In some embodiments, the left photodiodes PDL and right photodiodes PDR may be further isolated from each other by a junction isolation structure (not illustrated) or a combination of deep trench isolation structures included in isolation structure **304** and a junction isolation structure. In an embodiment, the junction isolation structure may be implemented by ion implantation of one or more dopant types (e.g., via P-type dopants such as boron) having an opposite conductivity type to the doped regions that form photodiodes included in pixel cell array **300** (e.g., via N-type dopants such as phosphorus or arsenic).

[0051] In some embodiments, a metal grid **316** is formed between individual color filters ("R", "G", or "B") of the color filter array. In the same or other embodiments, metal grid **316** may be vertically aligned with isolation structure **304** to form a plurality of apertures defining light exposure or sensing regions of individual pixels or pixel cells included in pixel cell array **300**. Apertures formed by metal grid **316** may be filled with individual color filters such that a given pixel or pixel cell is associated with a predetermined color (e.g., red, green, or blue). Metal grid **316** includes a plurality metal grid segments MG separating each individual color filter.

[0052] PDAF pixel cell **320** includes microlens **350-1** optically aligned with photosensitive region **309** to cover left photodiodes PDL and right photodiodes PDR. Metal shield **325** partially covers photosensitive region **309** and is further disposed between microlens **350-1** and left photodiodes PDL to block incident light (e.g., light **370**) from reaching left photodiodes PDL. However, it is appreciated that metal shield **325** may be arranged in a different orientation over right photodiodes PDR to block incident light from reaching right photodiodes PDR (e.g., such that two differently oriented PDAF pixel cells may be utilized to form left and right images for phase detection autofocus). Metal shield **325** has a lateral width W.sub.M that is less than lateral width W.sub.PDAF of PDAF pixel cell **320**. In the illustrated embodiment, microlens **350-1** is arranged to cover all or part of a light exposure region of PDAF pixel cell **320** depending on a position of PDAF pixel cell **320** within pixel cell array **300**. In some embodiments, metal shield **325** is formed at the same time as metal grid **316**. Metal grid **316** includes a plurality of metal grid segments MG, MG-**1** disposed between individual color filters (e.g., green color filter "G", blue color filter "B", red color filter "R").

[0053] PDAF pixel cell **320** further includes a spectral filter **340** (e.g., a red color filter) disposed between microlens **350-1** and metal shield **325**. In some embodiments, spectral filter **340** has a photoresponse different from the photoresponse of color filters laterally surrounding PDFAF pixel cell **320** (e.g., spectral filter **340** of PDAF pixel cell **320** corresponds to a red color filter as illustrated that is laterally surrounded by green or blue color filters). Put in another way, a first photoresponse of spectral filter **340** (e.g., a red color filter) included in PDAF pixel cell **320** is different than a second photoresponse of corresponding spectral filters including in imaging pixel cells (e.g., blue color filter or green color filter) laterally surrounding PDAF pixel cell **320** (e.g., imaging pixel cells **329**B, **331**G, **333**G, **335**B, etc.). Spectral filter **340** reduces crosstalk attributed to PDAF pixel cell **320** by reducing light reflecting from metal shield **325** and further attenuating light that adjacent color filters transmit. For example, incident light **370** may reflect off metal shield **325** to form reflected rays **370**R**1**, **370**R**2**, and/or **370**R**3**. However, spectral filter **340** filters or attenuates incident light **370** and/or reflected ray **370**R**1** that is outside of the red range (e.g., red light having a wavelength range between 600 nm and 750 nm is transmitted while other visible or non-visible light outside of the 600 nm to 750 nm range is attenuated). Consequently, reflected ray **370**R**1**, **370**R**2**, and **370**R**3** comprise primarily red light that is subsequently filtered or otherwise attenuated by the color filters of adjacent imaging pixel cells (e.g., blue or green color filters included in imaging pixel cells **329**B, **331**G, **333**G, and/or **335**B attenuate the primarily red light of reflected rays **370**R**1**, **370**R**2**, and **370**R**3** to reduce PD crosstalk).

[0054] To avoid lowering the selectivity of PDAF pixel cell **320**, spectral filter **340** does not fully cover metal shield **325**. Additionally, to increase quantum efficiency, no spectral filter is disposed over an uncovered portion of photosensitive region **309** defined by metal shield **325** (i.e., the partial coverage of photosensitive region **309** provided by metal shield **325** defines an uncovered portion of photosensitive region **309**). Thus, in some embodiments, spectral filter **340** does not extend over the uncovered portion of the photosensitive region **309**. Additionally, spectral filter **340** partially covers metal shield **325** to define an uncovered portion (e.g., corresponding to gap **341**) of metal shield **325**. Put in another way, spectral filter **340** partially covers metal shield **325** and metal shield **325** partially covers photosensitive region **309**. As illustrated, a first lateral width W.sub.S of spectral filter **340** is less than a second lateral width W.sub.M of metal shield **325** along a direction parallel to the backside surface BS to form gap **341** (e.g., corresponding to a portion of metal shield **325** that is not covered by spectral filter **340**) extending toward an edge **325**E of metal shield **325** when viewed from a plan view (e.g., as shown by plan view **300**-P illustrated in FIG. **3**). In the illustrated embodiment, a third lateral width W.sub.GAP of the gap **341** along the direction parallel to the backside surface BS is less than the first lateral width W.sub.S of spectral filter **340**. In some embodiments, gap **341** is filled by a transparent microlens material **330** included in microlens **350-1**. In the same or other embodiments, microlens material **330** fills a gap **319** defined by metal shield **325** such that microlens material **330** is disposed between metal shield **325** (e.g., edge **325**E) and a proximal metal grid segment MG-**1** included in metal grid **316**. In an embodiment, edge **325**E of metal shield **325** corresponds to an inner edge of metal shield **325** disposed closer to a midpoint of photosensitive region **309** (e.g., a lateral point disposed between left photodiodes PDL and right photodiodes PDR) relative to an outer edge (e.g., edge of metal shield **325** that interfaces or is disposed proximate to second spectral filter **344** that is opposite of edge **325**E).

[0055] In some embodiments, the color filter material (e.g., red color filter material) for forming spectral filter **340** is formed of a photo-resistive material or a resin material containing red color agent such as a red pigment or red dye that transmits or passes red light (e.g., light having wavelength ranging from 600 nm to 750 nm) while reflecting, absorbing, or otherwise attenuating other colors of light (e.g., light outside of the 600 nm to 750 nm range). In illustrated embodiments, microlens material **330** (e.g., a polymer transparent through the visible range of wavelengths) is disposed between metal shield **325** and proximal metal grid segment MG-**1** that separates microlens material **330** and adjacent color filters (e.g., spectral filter **342** corresponding to a green color filter) of a neighboring imaging pixel cell (e.g., QPD pixel cell **333**G). In the same or other embodiments, microlens material **330** is disposed between spectral filter **340** and adjacent or neighboring color filters (e.g., spectral filter **342**) of a neighboring imaging pixel cell (e.g. QPD pixel cell **333**G).

[0056] In some embodiments, spectral filter **340** has a first vertical thickness H**1** along a direction normal to the backside surface BS of semiconductor material **302** (e.g., depthwise direction). In some embodiments, first vertical thickness H**1** is substantially (e.g., does not deviate but for due to manufacturing or processing limitations such as 10% or less, 5% or less, or the like) uniform throughout the first lateral width W.sub.S of spectral filter **340**. In the same or other embodiments, metal shield **325** has a second vertical thickness H**2** along the direction normal to the backside surface BS. In some embodiments, a combined vertical thickness (e.g., thickness along the direction normal to the backside surface BS) of spectral filter **340** and metal shield **325** is the same as a third vertical thickness of an adjacent color filter included in an adjacent QPD pixel cell (e.g., third vertical thickness H**3** of second spectral filter **344** included in QPD pixel cell **331**G adjacent to PDAF pixel cell **320**). In some embodiments, a top surface **340**T of spectral filter **340** is flush with a top surface of an adjacent or interfacing color filter included in an adjacent QPD pixel cell (e.g., top surface **344**T of second spectral filter **344** is flush with spectral filter **340**) for processing consideration and to reduce the impact on sensitivity of the adjacent QPD pixel cells (e.g., QPD pixel cell **331**G). In a different embodiment, the combined vertical thickness of spectral filter **340** and metal shield **325** is greater than third vertical thickness of the adjacent color filter included in the adjacent QPD pixel cell (third vertical thickness H**3** of second spectral filter **344** included in QPD pixel cell **331**G adjacent to PDAF pixel cell **320**).

[0057] In the illustrated embodiment, spectral filter **340** extends laterally from edge **320**E of PDAF pixel cell **320** (e.g., corresponding to an outer edge of spectral filter **340**) a lateral width W.sub.S toward a center of PDAF pixel cell **320**. Spectral filter **340** extends from an edge **344**E of second spectral filter **344** (e.g., a green color filter corresponding to second spectral filter **344** adjacent to a red color filter corresponding to spectral filter **340**) toward a center of PDAF pixel cell **320**. In some embodiments, spectral filter **340** abuts second spectral filter **344** (e.g., spectral filter **340** directly contacts second spectral filter **344**).

[0058] In some embodiments, first lateral width W.sub.S of spectral filter **340** is less than second lateral width W.sub.M of metal shield **325** along the direction parallel to the backside surface Bs of semiconductor material **302** such that there is a gap between edge **340**E of spectral filter **340** and edge **325**E of metal shield **325** filled by microlens material **330**. In some embodiments, spectral filter **340** is sandwiched between second spectral filter **344** and microlens material **330** to minimize sensitivity or angular selectivity impact on right photodiodes PDR. Arranging a gap between edge **340**E of spectral filter **340** and edge **325**E of metal shield **325** may help to reduce unintentional attenuation of light that is directed to uncovered photodiodes (e.g., right photodiodes PDR) of PDAF pixel cell **320** (e.g., to avoid reducing sensitivity of image signals generated by PDAF pixel cell **320** that are used for phase detection autofocus). In some embodiments, a ratio between first lateral width W.sub.S of spectral filter **340** and second lateral width W.sub.M of metal shield **325** is from 0.6 to 1.

[0059] In some embodiments, metal shield **325** may cover up to 50% of photosensitive region **309** or PDAF pixel cell **320**. In such an embodiment, second lateral width W.sub.M of metal shield **325** along the direction parallel to the backside surface BS is half of width W.sub.PDAF of PDAF pixel cell **320**. In some embodiments, the spectral filter **340** has an area coverage amount ranging from 30% to 50% of photosensitive region **309** (e.g., an area defined by width W.sub.PDAF of PDAF pixel cell **320**). In some embodiments, spectral filter **340** for PDAF pixel cell **320** may vary in lateral width (e.g., coverage amount) depending on a location within pixel cell array **300** and/or based on a position or orientation of metal shield **325** (e.g., left or right orientation that respectively cover left photodiodes PDL or right photodiodes PDR) to accommodate for a chief ray angle for the lens employed by the image sensor (e.g., the chief ray angle for a given pixel or pixel cell changes based on its relative position within pixel cell array **300**).

[0060] In some embodiments, the dimension or coverage amount of spectral filter **340** may be configured based on target optical performance such as crosstalk level (e.g., acceptable level of crosstalk for second level pixels) and/or pixel sensitivity or angular selectivity for PDAF pixel cell **320**. More specifically, third lateral width W.sub.GAP of gap **341** may be held constant (e.g., substantially equal within limitations of manufacturing constraints) across pixel cell array **300** to improve pixel uniformity. In one embodiment, it may be preferable to limit a coverage amount of spectral filter **340** (e.g., to cover 30% of photosensitive region **309** defined by width W.sub.PDAF of PDAF pixel cell **320**) to avoid lowering angular selectivity of PDAF pixel cell **320**. In the same or other embodiments, spectral filter **340** partially covers metal shield **325** (e.g., such that there is third lateral width W.sub.GAP corresponding to a distance between edge **340**E of spectral filter **340** and edge **325**E of metal shield **325** along the direction parallel to the backside surface BS that is non-zero) to mitigate selectivity or sensitivity impact on PDAF pixel cell **320**. In some embodiments, third lateral width W.sub.GAP of gap **341** is configured to be at most about (e.g., within 10% or less) 0.4 times second lateral width W.sub.M of metal shield **325** such that spectral filter **340** provides sufficient coverage for filtering out or removing reflected light with short wavelengths (e.g., blue or green light) for improving crosstalk.

[0061] In embodiments where spectral filter **340** of PDAF pixel cell **320** corresponds to a red color filter and neighboring or adjacent color filters of imaging pixel cells correspond to green or blue color filters (e.g., second spectral filter **344** of QPD pixel cell **331**G corresponds to a green color filter), PDAF pixel cell **320** may reduce crosstalk of first, second, and third level pixels as well as corner pixels (see, e.g., FIG. **2**A and associated discussion defining pixel levels relative to a given PDAF pixel cell). As illustrated, spectral filter **340** placed between microlens **350-1** and metal shield **325** causes portions or all of light **370** that is reflected from metal shield **325** to first be filtered by spectral filter **340** (e.g., only portions of reflected light having wavelength between 600 nm to 750 nm is allowed to pass through spectral filter **340**). For example, reflected light **370**R**1** after propagating through spectral filter **340** would only have red light components as short light wavelength component (i.e., non-red light) is filtered or otherwise attenuated before reaching to neighboring imaging pixel cells (e.g., pixel cell **331**G and/or **329**B). The reflected light rays **370**R**2** and **370**R**3** generated from reflected light rays **370**R**1** propagating along a curvature of a microlens **350-2** subsequently enter respective color filters of neighboring imaging pixels (e.g., second spectral filter **344** corresponding to a green color filter and a blue color filter of QPD pixel cell **329**B) before reaching or absorbed by respective photodiodes. Since reflected light rays **370**R**1** only has red light components after being filtered by spectral filter **340** on metal shield **325**, reflected light rays **370**R**2** and **370**R**3** also only have red light components and thus would be attenuated or otherwise filtered by the green or blue color filters of the imaging pixel cells adjacent to PDAF pixel cell **320** (e.g., QPD pixel cells **329**B and/or **331**G). As such, PD crosstalk issues (e.g., crosstalk associated with second level pixels in particular) induced by metal shield **325** may be reduced.

[0062] FIG. **4**A illustrates an exemplary partition of pixel cell array **400** that includes a plurality of PDAF pixel cells **320**, in accordance with teachings of the disclosure. More specifically, pixel cell array **400** is an example of how lateral widths of metal shield **325** and/or spectral filter **340** may be varied based on a position of a given PDAF pixel cell within pixel cell array **400** to accommodate chief ray angle. More specifically, the magnitude of the chief ray angle associated with the given PDAF pixel cell increases as the position of the given PDAF pixel cell deviates from a center of pixel cell array **400**, which may impact the functionality of the given PDAF pixel cell (e.g., the asymmetry provided by the metal shield may be limited due to chief ray angle). Thus, in order to compensate or otherwise accommodate for chief ray angle, embodiments of the disclosure vary the lateral widths of the metal shield **325** and/or spectral filter **340** of the plurality of PDAF pixel cells **320** based on a position of a given PDAF pixel cell within pixel cell array **400**. In the illustrated embodiment, pixel cell array **400** is partitioned into 9 regions of interest (e.g., ROI **1** through ROI **9**) for the following discussion of placement and configuration of metal shield **325** and/or spectral filter **340** included in plurality of PDAF pixel cells **320** illustrated in FIG. **4**B-**4**E to accommodate chief ray angle.

[0063] Referring back to FIG. **4**A, ROI **1**, ROI **2**, and ROI **3** correspond to location in a left part of pixel cell array **400** with respect to a center of pixel cell array **400** where the chief ray angle is non-zero degrees in a negative direction. For example, ROI **1** may refer to a top-left corner of pixel cell array **400**. ROI **2** refers to center-left region of pixel cell array **400**. ROI **3** refers to bottom-left corner of pixel cell array **400**. ROI **4**, ROI **5** ROI **6** correspond to a center region of pixel cell array **400** where chief ray angle is 0°. ROI **7**, ROI **8**, and ROI **9** correspond to a right part of pixel cell array **400** with respect to the center of pixel cell array **400** where chief ray angle is non-zero degrees in a positive direction. In some embodiments, ROI **1**, ROI **2**, and ROI **3** correspond to a location in a leftmost part of pixel cell array **400** proximate to a perimeter boundary of pixel cell array **400** where the magnitude of the chief ray angle is the largest in the negative direction. In the same or other embodiments, ROI **7**, ROI **8**, and ROI **9** correspond to a location in the rightmost part of pixel cell array **400** proximate to a perimeter boundary of pixel cell array **400** where the magnitude of the chief ray angle is the largest in the positive direction.

[0064] FIG. **4**B illustrates a first pair of PDAF pixel cells including a left-oriented PDAF pixel cell **320**-PDAF-LA and a right-oriented PDAF pixel cell **320**-PDAF-RA located within any of ROI **1**, ROI **2**, or ROI **3** illustrated in FIG. **4**A, in accordance with teachings of the disclosure. In some embodiments, a metal shield (e.g., **325**-LA or **325**-RA illustrated in FIG. **4**E) and a spectral filter (e.g., **340**-LA or **340**-RA) included in PDAF pixel cells **320**-PDAF-LA and **320**-PDAF-RA may be configured to accommodate a chief ray angle for a lens employed by the image sensor (e.g., −33°).

[0065] FIG. **4**C illustrates a second pair of PDAF pixel cells including a left-oriented PDAF pixel cell **320**-PDAF-LB and a right-oriented PDAF pixel cell **320**-PDAF-RB located within any of ROI **4**, ROI **5**, or ROI **6** illustrated in FIG. **4**A, in accordance with teachings of the disclosure. In some embodiments, a metal shield (e.g., **325**-LB or **325**-RB illustrated in FIG. **4**E) and a spectral filter (e.g., **340**-LB or **340**-RB) included in PDAF pixel cells **320**-PDAF-LB and **320**-PDAF-RB may be configured to accommodate a chief ray angle for a lens employed by the image sensor (e.g., 0°).

[0066] FIG. **4**D illustrates a third pair of PDAF pixel cells including a left-oriented PDAF pixel cell **320**-PDAF-LC and a right-oriented PDAF pixel cell **320**-PDAF-RC located within any of ROI **7**, ROI **8**, or ROI **9** illustrated in FIG. **4**A, in accordance with teachings of the disclosure. In some embodiments, a metal shield (e.g., **325**-LC or **325**-RC illustrated in FIG. **4**E) and a spectral filter (e.g., **340**-LC or **340**-RC) included in PDAF pixel cells **320**-PDAF-LC and **320**-PDAF-RC may be configured to accommodate a chief ray angle for a lens employed by the image sensor (e.g., 33°).

[0067] FIG. **4**E illustrates a more detailed view of left and right oriented PDAF pixel cells illustrated in FIG. **4**B-**4**E, in accordance with teachings of the disclosure. Left oriented PDAF pixel cells include **320**-PDAF-LA, **320**-PDAF-LB, and **320**-PDAF-LC and right oriented PDAF pixel cells include **320**-PDAF-RA, **320**-PDAF-RB, and **320**-PDAF-RC, which may be variations of plurality of PDAF pixel cells **320** illustrated in FIG. **3**. Accordingly, it is appreciated that the left and right oriented PDAF pixel cells illustrated in FIG. **4**A-**4**E may have the same or similar features (e.g., descriptions of like-labeled features may similarly apply). For example, PDAF pixel cells **320**-PDAF-LA, **320**-PDAF-LB, and **320**-PDAF-LC respectively include metal shields (e.g., **325**-LA with lateral width W.sub.M-LA, **325**-LB with lateral width W.sub.M-LB, and **325**-LC with lateral width W.sub.M-LC) and spectral filters (e.g., **340**-LA with lateral width W.sub.S-LA, **340**-LB with lateral width W.sub.S-LB, and **340**-LC with lateral width W.sub.S-LC) comparable to metal shield **325** and spectral filter **340** illustrated in FIG. **3**. Similarly, PDAF pixel cells **320**-PDAF-RA, **320**-PDAF-RB, and **320**-PDAF-RC respectively include metal shields (e.g., **325**-RA with lateral width W.sub.M-RA, **325**-RB with lateral width W.sub.M-RB, and **325**-RC with lateral width W.sub.M-RC) and spectral filters (e.g., **340**-RA with lateral width W.sub.S-RA, **340**-RB with lateral width W.sub.S-RB, and **340**-RC with lateral width W.sub.S-RC) comparable to metal shield **325** and spectral filter **340** illustrated in FIG. **3**. It is further appreciated that lateral widths of uncovered portions of respective metal shields (e.g., gaps) resultant of partial coverage provided by the spectral filters of PDAF pixel cells **320**-PDAF-LA, **320**-PDAF-LB, **320**-PDAF-LC, **320**-PDAF-RA, **320**-PDAF-RB, and **320**-PDAF-RC are respectively illustrated as W.sub.GAP-LA, W.sub.GAP-LB, W.sub.GAP-LC, W.sub.GAP-RA, W.sub.GAP-RB, and W.sub.GAP-RC.

[0068] In reference to PDAF pixel cells **320**-PDAF-LA and **320**-PDAF-RC, metal shields **325**-LA and **325**-RC for light shielding may be enlarged (e.g., based on W.sub.M-LA and W.sub.M-RC) to cover more than 50% but less than 100% of a pixel region (e.g., defined by photodiodes PD**1**, PD**2**, PD**3**, and PD**4** such that two photodiodes in a same column are completely covered while two photodiodes in an adjacent column are only partially covered by the metal shield) to detect left (e.g., for **320**-PDAF-LA) or right (e.g., for **320**-PDAF-RC) image signal information for phase detection autofocus. In reference to PDAF pixel cells **320**-PDAF-RA and **320**-PDAF-LC, metal shields **325**-RA and **325**-LC for light shielding may be reduced (e.g., based on W.sub.M-RA and W.sub.M-LC) to cover more than 0% but less than 50% of a pixel region (e.g., defined by

photodiodes PD**1**, PD**2**, PD**3**, and PD**4** such that two photodiodes in a same column are only partially covered while two photodiodes in an adjacent column are not covered by the metal shield) to detect left (e.g., for **320**-PDAF-LC) or right (e.g., for **320**-PDAF-RA) image signal information for phase detection autofocus. In reference to PDAF pixel cells **320**-PDAF-LB and **320**-PDAF-RB, metal shields **325**-LB and **325**-RB for light shielding may cover (e.g., based on W.sub.M-LB and W.sub.M-RB) 50% of a pixel region (e.g., defined by photodiodes PD**1**, PD**2**, PD**3**, and PD**4** such that two photodiodes in a same column are completely covered while two photodiodes in an adjacent column are not covered by the metal shield) to detect left (e.g., for **320**-PDAF-LB) or right (e.g., for **320**-PDAF-RB) image signal information for phase detection autofocus.
[0069] It is appreciated that a pair of differently oriented (e.g., left and right oriented) PDAF pixel cells located in a same region of interest (e.g., pixel cell **320**-PDAF-LA and **320**-PDAF-RA, which may be collectively located in one of ROI **1**, ROI **2**, or ROI **3** illustrated in FIG. **4**A) may be used as an autofocus point for said region of interest or location (e.g., to determine whether said region of interest or location is in focus and/or how much lens displacement is needed to focus said region of interest or location). It is further appreciated that the pair of differently oriented PDAF pixel cells located in the same region of interest are separated from one another by at least one QPD imaging pixel cell as illustrated in FIG. **4**B-**4**D.
[0070] Referring back to FIG. **4**E, it is appreciated that in some embodiments, a relative amount of metal shields (e.g., **325**-LA, **325**-LB, **325**-LC, **325**-RA, **325**-RB, and **325**-RC) uncovered by spectral filters (e.g., **340**-LA, **340**-LB, **340**-LC, **340**-RA, **340**-RB, and **340**-RC) may be held constant (e.g., substantially equal within limitations of manufacturing constraints). In other words, in some embodiments third lateral widths (e.g., W.sub.GAP-LA, W.sub.GAP-LB, W.sub.GAP-LC, W.sub.GAP-RA, W.sub.GAP-RB, and W.sub.GAP-RC) is held constant across pixel cell array **400**. For example, as a first lateral width (e.g., W.sub.M-LA, W.sub.M-LB, W.sub.M-LC, W.sub.M-RA, W.sub.M-RB, and W.sub.M-RC) of the metal shields is adjusted based on an orientation and position of a given PDAF pixel cell within pixel cell array **400**, then a second lateral width (e.g., W.sub.S-LA, W.sub.S-LB, W.sub.S-LC, W.sub.S-RA, W.sub.S-RB, and W.sub.S-RC) of the spectral filters may also be adjusted while third lateral widths (e.g., W.sub.GAP-LA, W.sub.GAP-LB, W.sub.GAP-LC, W.sub.GAP-RA, W.sub.GAP-RB, and W.sub.GAP-RC) of gaps is held constant for all PDAF pixel cells included in pixel cell array **400**.
[0071] As illustrated in FIG. **4**E, PDAF pixel cells **320**-PDAF-LA, **320**-PDAF-LB, and **320**-PDAF-LC have a same orientation, but represent different positions within a pixel cell array and thus are associated with different chief ray angles. For example, **320**-PDAF-LB is representative of a chief ray angle (e.g., 0°) greater than a chief ray angle (e.g., −33°) associated with **320**-PDAF-LA but less than a chief ray angle (e.g., 33°) associated with **320**-PDAF-LC. In such an embodiment, a lateral width W.sub.M-LB of metal shield **325**-LB is greater than a lateral width W.sub.M-LC of metal shield **325**-LC but less than a lateral width W.sub.M-LA of metal shield **325**-LA. In the same or another embodiment, a lateral width W.sub.S-LB of spectral filter **340**-LB is greater than a lateral width W.sub.S-LC of spectral filter **340**-LC but less than a lateral width W.sub.S-LA of spectral filter **340**-LA. In the same or other embodiments, third lateral widths W.sub.GAP-LA, W.sub.GAP-LB, and W.sub.GAP-LC of uncovered portions of metal shields **325**-LA, **325**-LB, and **325**-LC is the same.
[0072] As illustrated in FIG. **4**E, PDAF pixel cells **320**-PDAF-RA, **320**-PDAF-RB, and **320**-PDAF-RC have a same orientation, but represent different positions within a pixel cell array and thus are associated with different chief ray angles. For example, **320**-PDAF-RB is representative of a chief ray angle (e.g., 0°) greater than a chief ray angle (e.g., −33°) associated with **320**-PDAF-RA but less than a chief ray angle (e.g., 33°) associated with **320**-PDAF-RC. In such an embodiment, a lateral width W.sub.M-RB of metal shield **325**-RB is less than a lateral width W.sub.M-RC of metal shield **325**-RC but greater than a lateral width W.sub.M-RA of metal shield **325**-RA. In the same or another embodiment, a lateral width W.sub.S-RB of spectral filter **340**-RB is less than a lateral

width W.sub.S-RC of spectral filter **340**-RC but greater than a lateral width W.sub.S-RA of spectral filter **340**-RA. In the same or other embodiments, third lateral widths W.sub.GAP-RA, W.sub.GAP-RB, and W.sub.GAP-RC of uncovered portions of metal shields **325**-RA, **325**-RB, and **325**-RC is the same. In some embodiments, paired PDAF pixel cells (e.g., a first pair including **320**-PDAF-LA and **320**-PDAF-RA, a second pair including **320**-PDAF-LB and **320**-PDAF-RB, and/or a third pair including **320**-PDAF-LC and **320**-PDAF-RC) have a same lateral width for uncovered portions of the respective metal shields.

[0073] In some embodiments, each individual pair of PDAF pixel cells (e.g., the first pair including **320**-PDAF-LA and **320**-PDAF-RA, the second pair including **320**-PDAF-LB and **320**-PDAF-RB, and/or the third pair including **320**-PDAF-LC and **320**-PDAF-RC) are each individually arranged to be laterally surrounded by imaging pixel cells (e.g., QPD pixel cells **310** illustrated in FIG. **4**B-**4**D, which may correspond to plurality of QPD pixel cells **210** illustrated in FIG. **2**A-**2**B). Groups of pixels or photodiodes included in QPD pixel cells **310** surrounding the first pair, second pair, and/or third pair of PDAF pixel cells may be classified as first level pixels (e.g., first level pixels **230** illustrated in FIG. **2**A-**2**B), corner pixels (e.g., corner pixels **240** illustrated in FIG. **2**A-**2**B), second level pixels (e.g., second level pixels **250** illustrated in FIG. **2**A-**2**B), or third level pixels (e.g., third level pixels **260** illustrated in FIG. **2**A-**2**B).

[0074] In some embodiments, spectral filter **340** (e.g., **340**-LA, **340**-LB, **340**-LC, **340**-RA, **340**-RB, and/or **340**-RC), which partially covers underlying metal shield **325** (e.g., **325**-LA, **325**LB, **325**-LC, **325**-RA, **325**-RB, and/or **325**-RC) may be shifted towards a center of a photosensitive region (e.g., where PD**1**, PD**2**, PD**3**, and PD**4** meet) of a given PDAF pixel cell to address color imbalance associated with first level pixels. For example, an edge of spectral filter **340** proximal to the center of the photosensitive region of the given PDAF pixel cell may be aligned with a corresponding edge of the metal shield **325** proximal to the center of the photosensitive region of the given PDAF pixel cell. In other words, edges of spectral filter **340** and metal shield **325** closest to a center or midpoint of a given PDAF pixel cell are aligned.

[0075] FIG. **5** illustrates a cross-sectional view **500**-X and a plan view **500**-P of a partial pixel cell array of an image sensor including a PDAF pixel cell **520** with reduced induced crosstalk, in accordance with teachings of the disclosure. PDAF pixel cell **520** represents an improved phase detection structure relative to PDAF pixel cell **220** illustrated in FIG. **2**A-**2**B. In the illustrated embodiment, PDAF pixel cell **520** includes a plurality of photodiodes, including left photodiodes PDL and right photodiodes PDR (e.g., first, second, third, and forth photodiodes arranged in a two-by-two array) surrounded by imaging pixels (e.g., pixel cell **529**B including photodiode PD**1** **562**, pixel cell **531**G including photodiode PD**2** **552** and PD**1** **532**, pixel cell **533**G including photodiode PD**2** **534** and PD**1** **554**, and pixel cell **535**B including photodiode **564**) formed within semiconductor material **502**. PDAF pixel cell **520** further includes metal shield **525**, spectral filter **540**, and microlens **550-1**.

[0076] PDAF pixel cell **520** includes a photosensitive region **509** (e.g., defined by isolation structure **504**) formed in semiconductor material **502** corresponding to regions where left photodiodes PDL and right photodiodes PDR are located (e.g., doped regions of semiconductor material **502** that collectively form left photodiodes PDL and right photodiodes PDR). Semiconductor material **502** includes front side surface FS and the backside surface BS opposite of the front side FS. Metal shield **525** is disposed between microlens **550-1** and photosensitive region **509** (e.g., to block incident light from reaching left photodiodes PDL or right photodiodes PDR depending on an orientation of PDAF pixel cell **520**). In the illustrated embodiment, pixel cell array **500** includes isolation structure **504** (e.g., a plurality of deep trench isolation structures) formed in semiconductors material **502** to provide isolation between individual photodiodes (e.g., left photodiodes PDL and right photodiodes PDR may be isolation from each other via isolation structure **504**) included in phase detection autofocus pixel cells and normal imaging pixel cells (e.g., pixel cells **529**B, **531**G, **533**G, **535**G, and so on). In some embodiments, a metal grid **516** is formed between individual color filters ("R", "G", or "B") of the color filter array. In the same or other embodiments, metal grid **516** may be vertically aligned with isolation structure **504** to form a plurality of apertures defining light exposure or sensing regions of individual pixels or pixel cells included in pixel cell array **500**. Apertures formed by metal grid **516** may be filled with individual color filters such that a given pixel or pixel cell is associated with a predetermined color (e.g., red, green, or blue). Metal grid **516** includes a plurality metal grid segments MG separating each individual color filter. QPD pixel cell **531**G includes second spectral filter **544** (e.g., a green color filter) with microlens **550-2** optically aligned with underlying photodiodes (i.e., a two-by-two array of photodiodes including PD**1 5332** and PD**2 552**). QPD pixel cell **533**G includes another spectral filter **542** (e.g., a green color filter) with a corresponding microlens optically aligned with underlying photodiodes (i.e., a two-by-two array of photodiodes including PD**1 554** and PD**2 534**).

[0077] PDAF pixel cell **520** includes many similar features as PDAF pixel cell **320** illustrated in FIG. **3** and thus may include like-labeled elements (e.g., semiconductor material **502** may correspond to semiconductor material **302**, isolation structure **504** may correspond to isolation structure **504**, microlens **550-1** may correspond to microlens **350-1**, and so on) and thus may have the same or similar features. One difference is spectral filter **540** is shifted towards a center of PDAF pixel cell **520** (e.g., based on lateral dimensions of photosensitive region **509** and/or a width W.sub.PDAF′ of PDAF pixel cell **520**) such that edge **540**E is aligned with an inner edge **525**E of metal shield **525**. In the illustrated embodiment of FIG. **5**, edge **525**E of metal shield **525** corresponds to an inner edge opposite of an outer edge of metal shield **525**. It is appreciated that the inner edge of metal shield (i.e., edge **525**E) is disposed closer to a center of PDAF pixel cell **520** (e.g., based on lateral dimensions of photosensitive region **509** and/or a width W.sub.PDAF′ of PDAF pixel cell **520**).

[0078] However, since spectral filter **540** still only partially cover metal shield **525** a portion of metal shield **525** remains uncovered by spectral filter **540** (e.g., proximate to an outer edge of metal shield **525**). In the illustrated embodiment, an adjacent color filter is configured to extend over and onto the uncovered portion of metal shield **525**. In other words, the gap (i.e., uncovered portion of metal shield **525** due to partial coverage provided by spectral filter **540**) is filled by a second spectral filter (e.g., spectral filter **544**) having a second photoresponse (e.g., a photoresponse associated with a green or blue color filter) different from a first photoresponse (e.g., photoresponse associated with a red color filter) of spectral filter **540**. In the illustrated embodiment, second spectral filter **544**, or more specifically, an extended portion **544**EXT of second spectral filter **544**, is disposed between microlens **550-1** and metal shield **525**. In some embodiments, extended portion **544**EXT of second spectral filter **544** is adjacent to spectral filter **540**. In the same or other embodiments, extended portion **544**EXT of second spectral filter **544** is flush or otherwise directly contacts spectral filter **540**.

[0079] In some embodiments, a corresponding lateral width W.sub.S2_EXT of second spectral filter **544** disposed over metal shield **525** (e.g., extended portion **544**EXT) along the direction parallel to the backside surface BS is less than a first lateral width W.sub.S′ of spectral filter **540**. In the same or other embodiments, a vertical thickness (e.g., H**1**) of spectral filter **540** is substantially (e.g., within manufacturing tolerances such as 10%) uniform throughout first lateral width W.sub.S′ of spectral filter **540**. In some embodiments, a vertical thickness of second spectral filter **544** along the direction parallel to the backside surface BS is not uniform (e.g., second spectral filter **544** has a vertical thickness of H**3** for a portion not disposed over metal shield **525** and a vertical thickness H**1** for extended portion **544**EXT disposed over metal shield **525**). In some embodiments, a pixel associated with PD**1 532** included in pixel cell **531**G corresponds to a first level pixel that is row or column adjacent to PDAF pixel cell **520**.

[0080] As illustrated in FIG. **5**, a position of spectral filter **540** is shifted toward a center of a pixel region (e.g., based on photosensitive region **509**) of PDAF pixel **520** and away from microlens **550-2** of the adjacent QPD pixel (e.g., pixel associated with PD**1 532** included in QPD pixel cell

541G) that is configured as a normal imaging pixel. Edge **540**E of spectral filter **540** may be aligned with edge **525**E of metal shield **525** used for blocking light (e.g., that would otherwise be incident on left photodiodes PDL). In some embodiments, edge **540**E of spectral filter **540** and edge **525**E of metal shield **525** may be vertically aligned with a centerline of a deep trench isolation structure included in isolation structure **504** that separates left photodiodes PDL and right photodiodes PDR of PDAF pixel cell **520** to reduce an impact on angular selectivity of PDAF pixel cell **520**.
[0081] Lateral width W.sub.S′ of spectral filter **540** may be less than lateral width W.sub.M′ of metal shield **525**. For example, lateral width W.sub.S′ of spectral filter **540** may be about 60% of lateral width W.sub.M′ of metal shield **525**. In some embodiments, lateral width W.sub.S′ of spectral filter **540** may range from about 60% to 100% of lateral width W.sub.M′ of metal shield **525**.
[0082] In the illustrated embodiment, second spectral filter **544** is disposed between microlens **550-2** and photodiodes (e.g., a two-by-two array of photodiodes including PD**1** **532** and PD**2** **552**) included in QPD pixel cell **531**G adjacent to PDAF pixel cell **520**. Second spectral filter **544** includes extended portion **544**EXT that extends over the pixel region (e.g., photosensitive region **509**) of PDAF pixel cell **520**. Extended portion **544**EXT of spectral filter **544** may be formed or otherwise disposed on metal shield **525**. That is, a portion of second spectral filter **544** (e.g., extended portion **544**EXT) is formed between microlens **550-1** and metal shield **525**. Put in another way, second spectral filter **544** includes a first portion formed or otherwise disposed between microlens **550-2** and photodiodes included in QPD pixel cell **531**G adjacent to PDAF pixel cell **520** and a second portion formed or otherwise disposed between microlens **550-1** and photodiodes included in PDAF pixel cell **520** (e.g., left photodiodes PDL) to increase light sensitivity of the adjacent QPD pixel cell (e.g., QPD pixel cell **531**G) and further reduce PD crosstalk (e.g., crosstalk included by PDAF pixel cell **520**).
[0083] In some embodiments, spectral filter **540** is disposed on metal shield **525** between extended portion **544**EXT of second spectral filter **5444** and microlens material **530** (e.g., a transparent polymer) of microlens **550-1**. In the same or other embodiments, microlens material **530** is disposed between metal shield **525** and a metal grid segment MG-**1**. In some embodiments, microlens material **530** is further disposed between the metal shield **525** and another spectral filter **542** included in another adjacent QPD pixel cell (e.g., QPD pixel cell **533**G such that QPD pixel cell **531**G, PDAF pixel cell **520**, and QPD pixel c ell **533**G are disposed along a same row or column) positioned on a right side of PDAF pixel cell **520** (e.g., closer to right photodiodes PDR relative to left photodiodes PDL).
[0084] In an embodiment, extended portion **544**EXT of second spectral filter **544** has a lateral width W.sub.S2_EXT that is less than a lateral width W.sub.M of metal shield **525** in the direction parallel to the backside surface BS of semiconductor material **502**. In some embodiments, lateral width W.sub.S2_EXT of extended portion **544**EXT is greater than 0 but less than lateral width W.sub.S′ of spectral filter **540**. In some embodiments, a combination of lateral width W.sub.S2_EXT of extended portion **544**EXT and lateral width W.sub.S of spectral filter **540** is substantially (e.g., within manufacturing tolerances such as 10%) the same as lateral width W.sub.M of metal shield **525**. In some embodiments, lateral width W.sub.S2_EXT of extended portion **544**EXT may be kept constant (e.g., similar to uncovered portion of metal shield **325** illustrated in FIG. **4**E). In some embodiments, width W.sub.S2_EXT of extended portion **544**EXT may vary or is otherwise based on lateral width W.sub.S of spectral filter **540**.
[0085] In some embodiments, extended portion **544**EXT of second spectral filter **544** may cover an area of photosensitive region **509** of PDAF pixel cell **520** from 1% to 20%. In some embodiments, lateral width W.sub.S2_EXT of extended portion **544**EXT of second spectral filter **544** is greater than 0 but less than 0.4 times lateral width W.sub.M′ of metal shield **525** while lateral width W.sub.S of spectral filter **540** is from 0.6 to 0.9 times lateral width W.sub.M′ of metal shield **525**. In

some embodiments, lateral width W.sub.S2_EXT of extended portion **544**EXT and lateral width W.sub.S′ of spectral filter **540** may be configured based on a width W.sub.PDAF′ of PDAF pixel cell **520**.

[0086] In operation, extended portion **544**EXT allows a portion of light directed through microlens **550-1** to be incident on or otherwise directed to photodiodes (e.g., first level pixel such as pixel associated with PD**1** **532**) included in the QPD pixel cell adjacent to PDAF pixel cell **520** (e.g., QPD pixel cell **531**G). In such a manner, PD crosstalk may be reduced and pixel uniformity may be improved. In some embodiments, second spectral filter **544** including extended portion **544**EXT may be formed of a green color filter material. In one example, the green color filter material is formed a photo-resistive material or a resin material containing green color agent such as green pigment or green dye) that transmits or passes green light (e.g., light having wavelength ranging from 500 nm to 570 nm) while reflecting and/or absorbing other colors of light. In the same or different embodiments, spectral filter **540** may be formed of a red color filter material. In one example, the red color filter material is formed of a photo-resistive material or a resin material containing red color agent such as red pigment or red dye) that transmits or passes red light (e.g., light having wavelength ranging from 600 nm to 750 nm) while reflecting and/or absorbing other colors of light.

[0087] In some embodiments, second spectral filter **544** including extended portion **544**EXT may be formed in the same process as green color filters for other “green” QPD pixels sharing a common mask. The formation of second spectral filter **544** including extended portion **544**EXT may occur after the formation of metal shield **525** but prior to deposition of microlens material **530** for forming a microlens array including microlens **550-1** and **550-2**. In some embodiments, formation of second spectral filter **544** including extended portion **544**EXT may occur before or after formation of spectral filter **540**.

[0088] FIG. **6** illustrates left and right oriented PDAF pixel cells with two spectral filters disposed over corresponding metal shields, in accordance with teachings of the disclosure. Left oriented PDAF pixel cells include **520**-PDAF-LA, **520**-PDAF-LB, and **520**-PDAF-LC and right oriented PDAF pixel cells include **520**-PDAF-RA, **520**-PDAF-RB, and **520**-PDAF-RC, which may be variations of plurality of PDAF pixel cells **520** illustrated in FIG. **5** and disposed across a pixel array in similar manner as illustrated in FIG. **4**B-**4**D. Accordingly, it is appreciated that the left and right oriented PDAF pixel cells illustrated in FIG. **6** may have the same or similar features (e.g., descriptions of like-labeled features may similarly apply). For example, PDAF pixel cells **520**-PDAF-LA, **520**-PDAF-LB, and **520**-PDAF-LC respectively include metal shields (e.g., **525**-LA with lateral width W.sub.M′-LA, **525**-LB with lateral width W.sub.M′-LB, and **525**-LC with lateral width W.sub.M′-LC) and spectral filters (e.g., **540**-LA with lateral width W.sub.S′-LA, **540**-LB with lateral width W.sub.S′-LB, and **540**-LC with lateral width W.sub.S′-LC) comparable to metal shield **525** and spectral filter **540** illustrated in FIG. **5**. Similarly, PDAF pixel cells **520**-PDAF-RA, **520**-PDAF-RB, and **520**-PDAF-RC respectively include metal shields (e.g., **525**-RA with lateral width W.sub.M′-RA, **525**-RB with lateral width W.sub.M′-RB, and **525**-RC with lateral width W.sub.M′-RC) and spectral filters (e.g., **540**-RA with lateral width W.sub.S′-RA, **540**-RB with lateral width W.sub.S′-RB, and **540**-RC with lateral width W.sub.S′-RC) comparable to metal shield **525** and spectral filter **540** illustrated in FIG. **5**. It is further appreciated that lateral widths of extended portions (e.g., **544**EXT-LA, **544**EXT-LB, **544**EXT-LC, **544**EXT-RA, **544**EXT-RB, and **544**EXT-RC) of second spectral filter **544** having respective lateral widths (e.g., **544**.sub.S2_EXT-LA, **544**.sub.S2_EXT-LB, **544**.sub.S2_EXT-LC, **544**.sub.S2_EXT-RA, **544**.sub.S2_EXT-RB, and **544**.sub.S2_EXT-RC) are comparable to extended portion **544**EXT of FIG. **5**.

[0089] It is appreciated that left oriented PDAF pixel cells **520**-PDAF-LA, **520**-PDAF-LB, and **520**-PDAF-LC and right oriented PDAF pixel cells **520**-PDAF-RA, **520**-PDAF-RB, and **520**-PDAF-RC are representative of PDAF pixel cells configured based on an orientation and position within a pixel cell array. For example, in reference to FIG. **4**A, PDAF pixel cells **520**-PDAF-LA and **520**-PDAF-RA may be positioned within ROI **1**, ROI **2**, or ROI **3**. In another example, PDAF pixel cells **520**-PDAF-LB and **520**-PDAF-RB may be positioned within ROI **4**, ROI **5**, or ROI **6**. In another example PDAF pixel cells **520**-PDAF-LC and **520**-PDAF-RC may be positioned within ROI **7**, ROI **8**, or ROI **9**. Put in another way, PDAF pixel cells **520**-PDAF-LB and **520**-PDAF-RB are associated with a chief ray angle (e.g., 0°) greater than a chief ray angle associated with PDAF pixel cells **520**-PDAF-LA and **520**-PDAF-RA (e.g., −33°) but less than a chief ray angle associated with PDAF pixel cells **520**-PDAF-LC and **520**-PDAF-RC (e.g., 33°).

[0090] In reference to PDAF pixel cells **520**-PDAF-LA and **520**-PDAF-RC, metal shields **525**-LA and **525**-RC for light shielding may be enlarged (e.g., based on W.sub.M′-LA and W.sub.M′-RC) to cover more than 50% but less than 100% of a pixel region (e.g., defined by photodiodes PD**1**, PD**2**, PD**3**, and PD**4** such that two photodiodes in a same column are completely covered while two photodiodes in an adjacent column are only partially covered by the metal shield) to detect left (e.g., for **520**-PDAF-LA) or right (e.g., for **520**-PDAF-RC) image signal information for phase detection autofocus. In reference to PDAF pixel cells **520**-PDAF-RA and **520**-PDAF-LC, metal shields **525**-RA and **525**-LC for light shielding may be reduced (e.g., based on W.sub.M′-RA and W.sub.M′-LC) to cover more than 0% but less than 50% of a pixel region (e.g., defined by photodiodes PD**1**, PD**2**, PD**3**, and PD**4** such that two photodiodes in a same column are only partially covered while two photodiodes in an adjacent column are not covered by the metal shield) to detect left (e.g., for **520**-PDAF-LC) or right (e.g., for **520**-PDAF-RA) image signal information for phase detection autofocus. In reference to PDAF pixel cells **520**-PDAF-LB and **520**-PDAF-RB, metal shields **525**-LB and **525**-RB for light shielding may cover (e.g., based on W.sub.M′-LB and W.sub.M′-RB) 50% of a pixel region (e.g., defined by photodiodes PD**1**, PD**2**, PD**3**, and PD**4** such that two photodiodes in a same column are completely covered while two photodiodes in an adjacent column are not covered by the metal shield) to detect left (e.g., for **520**-PDAF-LB) or right (e.g., for **520**-PDAF-RB) image signal information for phase detection autofocus. In one embodiment, spectral filters **540**-LB and **540**-RB may be configured to cover about (e.g., within 10% or otherwise based on manufacturing tolerances) 60% of an area of metal shields **525**-LB and **525**-RB, respectively, and extended portions **544**EXT-LB and **544**EXT-RB are configured to cover the remaining 40% area of metal shields **525**-LB and **525**-RB, respectively.

[0091] It is appreciated that a pair of differently oriented (e.g., left and right oriented) PDAF pixel cells located in a same region of interest (e.g., pixel cell **520**-PDAF-LA and **520**-PDAF-RA, which may be collectively located in one of ROI **1**, ROI **2**, or ROI **3**) may be used as an autofocus point for said region of interest or location (e.g., to determine whether said region of interest or location is in focus and/or how much lens displacement is needed to focus said region of interest or location). It is further appreciated that the pair of differently oriented PDAF pixel cells located in the same region of interest are separated from one another by at least one QPD imaging pixel cell.

[0092] In some embodiments, spectral filters, extended portions of second spectral filters, and metal shields for pairs of PDAF pixel cells (e.g., a first pair including **520**-PDAF-LA and **520**-PDAF-RA, a second pair including **520**-PDAF-LB and **520**-PBAR-RB, and a third pair including **520**-PDAF-LC and **520**-PDAF-RC) may be configured or arranged based on an incident angle of light desired for an adjacent QPD pixel (e.g., first level pixel such as pixel associated with PD**1** **532** of pixel cell **531**G illustrated in FIG. **5**) to receive.

[0093] Referring to FIG. **6**, it is appreciated that in some embodiments, a relative coverage amount of metal shields (e.g., **525**-LA, **525**-LB, **525**-LC, **525**-RA, **525**-RB, and **525**-RC) provided by extended portions (e.g., **544**EXT-LA, **544**EXT-LB, **544**EXT-LC, **544**EXT-RA, **544**EXT-RB, and **544**EXT-RC) of second spectral filters may be held constant (e.g., substantially equal within limitations of manufacturing constraints) for each PDAF pixel cell included in pixel cell array **500**. For example, first lateral widths (e.g., W.sub.M′-LA, W.sub.M′-LB, W.sub.M′-LC, W.sub.M′-RA, W.sub.M′-RB, and W.sub.M′-RC) of metal shields and second lateral widths (e.g., W.sub.S′-LA, W.sub.S′-LB, W.sub.S′-LC, W.sub.S′-RA, W.sub.S′-RB, and W.sub.S′-RC) of spectral filters may be adjusted based on an orientation and position of a given PDAF pixel cell (e.g., to accommodate chief ray angle) within pixel cell array **500** while third lateral widths (e.g., **544**.sub.S2_EXT-LA, **544**.sub.S2_EXT-LB, **544**.sub.S2_EXT-LC, **544**.sub.S2_EXT-RA, **544**.sub.S2_EXT-RB, and **544**.sub.S2_EXT-RC) of extended portions (e.g., **544**EXT-LA, **544**EXT-LB, **544**EXT-LC, **544**EXT-RA, **544**EXT-RB, and **544**EXT-RC) of second spectral filters are held constant for each PDAF pixel cell included in pixel cell array **500**.

[0094] As illustrated, PDAF pixel cells **520**-PDAF-RA, **520**-PDAF-RB, and **520**-PDAF-RC have a same orientation, but represent different positions within a pixel cell array and thus are associated with different chief ray angles. For example, **520**-PDAF-RB is representative of a chief ray angle (e.g., 0°) greater than a chief ray angle (e.g., −33°) associated with **520**-PDAF-RA but less than a chief ray angle (e.g., 33°) associated with **520**-PDAF-RC. In such an embodiment, a lateral width W.sub.M′-RB of metal shield **525**-RB is less than a lateral width W.sub.M′-RC of metal shield **525**-RC but greater than a lateral width W.sub.M′-RA of metal shield **525**-RA. In the same or another embodiment, a lateral width W.sub.S′-RB of spectral filter **540**-RB is less than a lateral width W.sub.S′-RC of spectral filter **540**-RC but greater than a lateral width W.sub.S′-RA of spectral filter **540**-RA. In the same or other embodiments, third lateral widths W.sub.S2_EXT-RA, W.sub.S2_EXT-RB, and W.sub.S2_EXT-RC of extended portions of second spectral filters **344** are the same. In some embodiments, paired PDAF pixel cells (e.g., a first pair including **520**-PDAF-LA and **520**-PDAF-RA, a second pair including **520**-PDAF-LB and **520**-PDAF-RB, and/or a third pair including **520**-PDAF-LC and **520**-PDAF-RC) have third lateral widths of extended portions of second spectral filters **344** that are the same.

[0095] In some embodiments, each individual pair of PDAF pixel cells (e.g., the first pair including **520**-PDAF-LA and **520**-PDAF-RA, the second pair including **520**-PDAF-LB and **520**-PDAF-RB, and/or the third pair including **520**-PDAF-LC and **520**-PDAF-RC) are each individually arranged to be laterally surrounded by imaging pixel cells (e.g., QPD pixel cells **310** illustrated in FIG. **4**B-**4**D, which may correspond to plurality of QPD pixel cells **210** illustrated in FIG. **2**A-**2**B). Groups of pixels or photodiodes included in QPD pixel cells **310** surrounding the first pair, second pair, and/or third pair of PDAF pixel cells may be classified as first level pixels (e.g., first level pixels **230** illustrated in FIG. **2**A-**2**B), corner pixels (e.g., corner pixels **240** illustrated in FIG. **2**A-**2**B), second level pixels (e.g., second level pixels **250** illustrated in FIG. **2**A-**2**B), or third level pixels (e.g., third level pixels **260** illustrated in FIG. **2**A-**2**B).

[0096] FIG. **7** illustrates a cross-sectional view **700**-X and a plan view **700**-P of a partial pixel cell array of an image sensor including a PDAF pixel cell **720** with a combination of a spectral filter and an antireflection layer for reduced induced crosstalk, in accordance with teachings of the disclosure. PDAF pixel cell **720** represents an improved phase detection structure relative to PDAF pixel cell **220** illustrated in FIG. **2**A-**2**B. In the illustrated embodiment, PDAF pixel cell **720** includes a plurality of photodiodes, including left photodiodes PDL and right photodiodes PDR (e.g., first, second, third, and forth photodiodes arranged in a two-by-two array) surrounded by imaging pixels (e.g., pixel cell **729**B including photodiode PD**1 762**, pixel cell **731**G including photodiode PD**2 752** and PD**1 732**, pixel cell **733**G including photodiode PD**2 734** and PD**1 754**, and pixel cell **735**B including photodiode **764**) formed within semiconductor material **702**. PDAF pixel cell **720** further includes metal shield **725**, spectral filter **740**, extended portion **744**EXT of second spectral filter **744**, antireflection layer **723**, and microlens **750-1**. In the illustrated embodiment, antireflection layer **723** is disposed between spectral filter **740** and metal shield **725**. In the same embodiment, antireflection layer **723** is disposed between extended portion **744**EXT of second spectral filter **744** and metal shield **725**.

[0097] PDAF pixel cell **720** includes a photosensitive region **709** (e.g., defined by isolation structure **704**) formed in semiconductor material **702** corresponding to regions where left photodiodes PDL and right photodiodes PDR are located (e.g., doped regions of semiconductor material **702** that collectively form left photodiodes PDL and right photodiodes PDR).

Semiconductor material **702** includes front side surface FS and the backside surface BS opposite of the front side FS. Metal shield **725** is disposed between microlens **750-1** and photosensitive region **709** (e.g., to block incident light from reaching left photodiodes PDL or right photodiodes PDR depending on an orientation of PDAF pixel cell **720**). In the illustrated embodiment, pixel cell array **700** includes isolation structure **704** (e.g., a plurality of deep trench isolation structures) formed in semiconductors material **702** to provide isolation between individual photodiodes (e.g., left photodiodes PDL and right photodiodes PDR may be isolation from each other via isolation structure **704**) included in phase detection autofocus pixel cells and normal imaging pixel cells (e.g., pixel cells **729**B, **731**G, **733**G, **735**G, and so on). In some embodiments, a metal grid **716** is formed between individual color filters ("R", "G", or "B") of the color filter array. In the same or other embodiments, metal grid **716** may be vertically aligned with isolation structure **704** to form a plurality of apertures defining light exposure or sensing regions of individual pixels or pixel cells included in pixel cell array **700**. Apertures formed by metal grid **716** may be filled with individual color filters such that a given pixel or pixel cell is associated with a predetermined color (e.g., red, green, or blue). Metal grid **716** includes a plurality metal grid segments MG separating each individual color filter. QPD pixel cell **731**G includes second spectral filter **744** (e.g., a green color filter) with microlens **750-2** optically aligned with underlying photodiodes (i.e., a two-by-two array of photodiodes including PD**1** **732** and PD**2** **752**). QPD pixel cell **733**G includes another spectral filter **742** (e.g., a green color filter) with a corresponding microlens optically aligned with underlying photodiodes (i.e., a two-by-two array of photodiodes including PD**1** **754** and PD**2** **734**).

[0098] PDAF pixel cell **720** includes many similar features as PDAF pixel cell **520** illustrated in FIG. **5** and PDAF pixel cell **320** illustrated in FIG. **3** and thus may include like-labeled elements (e.g., semiconductor material **702** may correspond to semiconductor material **502**, isolation structure **704** may correspond to isolation structure **504**, microlens **750-1** may correspond to microlens **550-1**, and so on) and thus may have the same or similar features. One difference is spectral filter **540** is antireflection layer **723** is formed over metal grid segments MG-**1**, MG, and metal shield **725** to further reduce crosstalk induced by metal shield **725** and/or metal grid segments included in metal grid **716**.

[0099] In the illustrated embodiment, antireflection layer **723** is disposed between microlens **750-1** and metal shield **725**. Antireflection layer **723** is further disposed between spectral filter **740** and metal shield **725** for reducing and/or attenuating reflection from metal shield **725** (e.g., reduce intensity or strength of reflected light **770**R**1** and subsequently reflected light **770**R**2** and **770**R**3**). In some embodiments, spectral filter **740** is configured to pass a light wavelength range that differs from or is outside of a transmission range of an adjacent color filter (e.g., second spectral filter **740** of adjacent QPD pixel cell **731**G and/or spectral filter **742** of adjacent QPD pixel cell **733**G).

[0100] In some embodiments, spectral filter **740** is a non-green color filter. In the same or different embodiments, spectral filter **740** may be configured to block light having a wavelength in a range from 500 nm to 550 nm. In some embodiments, spectral filter **740** may be a red color filter or a spectral filter that allows light with a wavelength range from 600 nm to 750 nm to propagate). In the same or different embodiments, red color filter material for forming spectral filter **740** is formed of a photo-resistive material or a resin material containing red color agent such as red pigment or red dye) that transmits or passes red light (e.g., light having a wavelength ranging from 600 nm to 750 nm) while reflecting and/or absorbing other colors of light.

[0101] In some embodiments, spectral filter **740** may be a blue color filter or a spectral filter that allows light with a wavelength range from 400 nm to 450 nm). Thus in some embodiments, the spectral filter **740** may be formed of a blue color filter material. In one example, the blue color filter material is formed of a photo-resistive material or a resin material containing a blue color agent such as blue pigment or blue dye) that transmits or passes blue light (e.g., light having a wavelength ranging from 400 nm to 450 nm) while reflecting and/or absorbing other colors of light.

[0102] In some embodiments, antireflection layer **723** is formed of titanium nitride (TiN) having a thickness ranging from 500 angstroms to 1000 angstroms. It is appreciated in other embodiments, antireflection layer **723** may be formed of other suitable materials such as titanium (Ti), a combination of titanium nitride and titanium, tungsten, or combinations thereof with appropriate thickness.
[0103] In operation, light **770** reflected by the metal shield **725** (e.g., reflected light ray **770**R**1** directed by microlens **550-1**) is first attenuated or reduced, for example, by antireflective layer **723** and then filtered by spectral filter **740** before propagating to an adjacent QPD pixel (e.g., pixel associated with PD**1** **732** included in QPD pixel cell **731**G). Spectral filter **740** is configured to filter portions of light included in reflected light ray **770**R**1** that is not filtered by the spectral filter (e.g., second spectral filter **744**) of the adjacent QPD pixel. As such, crosstalk for second level pixels (e.g., pixels associated with photodiode **752** PD**2** of QPD pixel cell **731**G) and third level pixels (e.g., photodiode **762** PD**1** of QPD pixel cell **729**B) from reflected light (e.g., reflected light ray **770**R**2** and/or **770**R**3**) may be reduced or minimized to improve PD induced crosstalk (e.g., due to PDAF pixel cell **720**).
[0104] In some embodiments, second spectral filter **744** of the adjacent QPD pixel cell **731**G may optionally have an extended portion **744**EXT that extends into a pixel region of PDAF pixel cell **720** in a similar manner to PDAF pixel cell **520** illustrated in FIG. **5**. In such an embodiment, extended portion **744**EXT is disposed between microlens material **730** and antireflection layer **723**. In some embodiments, extended portion **744**EXT has a lateral width W.sub.S2_EXT′ along a direction that is parallel to a backside surface BS of semiconductor material **702**. In some embodiments, a ratio of lateral width W.sub.S2_EXT′ of extended portion **744**EXT to lateral width W.sub.M″ of metal shield **725** may range from 0 to 0.4. In some embodiments, a combined lateral width of extended portion **744**EXT and the spectral filter **740** may be substantially (e.g., within manufacturing tolerances such as 10% the same as lateral width W.sub.M″ of metal shield **725**.
[0105] In some embodiments, antireflection layer **723** may optionally also be formed on or otherwise disposed on metal grid segments included in a metal grid **716**. In some embodiments, metal grid **716** and metal shield **725** may be formed during a same process.
[0106] FIG. **8** illustrates a block diagram of an imaging system **800** for an image sensor, in accordance with teachings of the disclosure. Imaging system **800** includes pixel cell array **805**, control circuitry **821**, readout circuitry **811**, and function logic **815**. In one embodiment, pixel cell array **805** is a two-dimensional (2D) array of QPD pixel cells (e.g., pixel cells P**1**, P**2** . . . , Pn) including imaging pixel cells and sparsely distributed PDAF pixel cells as illustrated in embodiments of the disclosure. As illustrated, pixel cells are arranged into rows (e.g., rows R**1** to Ry) and columns (e.g., column C**1** to Cx) to acquire image data of a person, place, object, etc., which can then be used to render an image or video of the person, place, object, etc. However, photodiodes, pixels, and/or pixel cells of imaging system **800** do not have to be arranged into rows and columns and may take other configurations.
[0107] In the various examples, readout circuitry **811** may be configured to read out the image signals at different conversion gains through column bitlines. In various examples, readout circuitry **811** may include current sources, routing circuitry, and comparators that may be included in analog to digital converters or otherwise.
[0108] In one embodiment, after each photodiode/pixel in pixel cell array **805** has acquired its image data or image charge, the image data is readout by readout circuitry **811** and then transferred to function logic **815**. In various examples, readout circuitry **811** may include amplification circuitry, analog-to-digital (ADC) conversion circuitry, or otherwise. Function logic **815** may simply store the image data or even manipulate the image data by applying post image effects (e.g., autofocus based on phase detection signal provided by PDAF pixel cells, crop, rotate, remove red eye, adjust brightness, adjust contrast, or otherwise). In the same or another embodiment, readout circuitry **811** may readout a row of image data at a time along readout column lines (illustrated) or

may readout the image data using a variety of other techniques (not illustrated), such as a serial readout or a full parallel readout of all pixels simultaneously. In one embodiment, control circuitry **821** is coupled to pixel cell array **805** to control operation pixel cell array **805**. For example, control circuitry **821** may generate a shutter signal for controlling image acquisition. In some embodiments, control circuitry **821** may be configured to generate drive signals (e.g., transfer signals, reset signals, and row-select signals) for controlling operation of pixel circuitries associated with pixel cell array **805**.

[0109] It is appreciated that imaging system **800** may be included in an image sensor that can be incorporated into digital camera, cell phone, laptop computer, automobile, surveillance camera or the like. Additionally, imaging system **800** may be coupled to other pieces of hardware such as a processor (general purpose or otherwise), memory elements, output (USB port, wireless transmitter, HDMI port, etc.), lighting/flash, electrical input (keyboard, touch display, track pad, mouse, microphone, etc.), and/or display. Other pieces of hardware may deliver instructions to imaging system **800**, extract image data from imaging system **800**, or manipulate image data supplied by imaging system **800**.

[0110] It is appreciated that embodiments of the disclosure illustrated in FIG. **1-8** may be fabricated using conventional semiconductor device processing and microfabrication techniques known by one of ordinary skill in the art, which may include, but is not limited to, photolithography, ion implantation, chemical vapor deposition, physical vapor deposition, thermal evaporation, sputter deposition, reactive-ion etching, plasma etching, wafer bonding, chemical mechanical planarization, and the like. It is appreciated that the described techniques are merely demonstrative and not exhaustive and that other techniques may be utilized to fabricate one or more components of various embodiments of the disclosure.

[0111] The above description of illustrated examples of the invention, including what is described in the Abstract, is not intended to be exhaustive or to limit the invention to the precise forms disclosed. While specific examples of the invention are described herein for illustrative purposes, various modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize.

[0112] These modifications can be made to the invention in light of the above detailed description. The terms used in the following claims should not be construed to limit the invention to the specific examples disclosed in the specification. Rather, the scope of the invention is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation.

Claims

1. A phase detection autofocus (PDAF) pixel cell, comprising: a photosensitive region including one or more photodiodes formed within a semiconductor material proximate to a backside surface of the semiconductor material; a microlens optically aligned with the photosensitive region; a metal shield disposed between the microlens and the photosensitive region; and a spectral filter disposed between the microlens and the metal shield, wherein a first lateral width of the spectral filter is less than a second lateral width of the metal shield along a direction parallel to the backside surface to form a gap extending from an edge of the spectral filter toward an edge of the metal shield when viewed from a plan view.

2. The PDAF pixel cell of claim 1, wherein the metal shield partially covers the photosensitive region, and wherein the spectral filter partially covers the metal shield.

3. The PDAF pixel cell of claim 2, wherein partial coverage of the photosensitive region provided by the metal shield defines an uncovered portion of the photosensitive region, and wherein the spectral filter does not extend over the uncovered portion of the photosensitive region.

4. The PDAF pixel cell of claim 1, wherein a third lateral width of the gap along the direction

parallel to the backside surface is less than the first lateral width of the spectral filter.
5. The PDAF pixel cell of claim 1, wherein the edge of the metal shield corresponds to an inner edge of the metal shield disposed closer to a midpoint of the photosensitive region relative to an outer edge of the metal shield, wherein the inner edge is opposite the outer edge.
6. The PDAF pixel cell of claim 5, wherein the gap is filled by microlens material included in the microlens.
7. The PDAF pixel cell of claim 1, wherein the edge of the metal shield corresponds to an outer edge of the metal shield, wherein the metal shield further includes an inner edge opposite the outer edge, and wherein the inner edge of the metal shield is disposed closer to a midpoint of the photosensitive region relative to the outer edge.
8. The PDAF pixel cell of claim 1, further comprising a second spectral filter disposed between the metal shield and the microlens adjacent to the spectral filter, wherein a first photoresponse of the spectral filter is different from a second photoresponse of the second spectral filter, and wherein the gap is filled by the second spectral filter.
9. The PDAF pixel cell of claim 8, wherein the spectral filter corresponds to a red color filter, and wherein the second spectral filter corresponds to a green color filter or a blue color filter.
10. The PDAF pixel cell of claim 8, wherein a corresponding lateral width of the second spectral filter disposed over the metal shield along the direction parallel to the backside surface is less than the first lateral width of the spectral filter.
11. The PDAF pixel cell of claim 1, further comprising an antireflection layer disposed between the spectral filter and the metal shield.
12. The PDAF pixel cell of claim 1, wherein a vertical thickness of the spectral filter along a direction normal to the backside surface is substantially uniform throughout the first lateral width of the spectral filter.
13. An image sensor, comprising: a pixel cell array including a plurality of pixel cells arranged in rows and columns, wherein the plurality of pixel cells includes phase detection autofocus (PDAF) pixel cells and imaging pixel cells laterally surrounding individual PDAF pixel cells included in the plurality of PDAF pixel cells, wherein the individual PDAF pixel cells each includes: a photosensitive region including one or more photodiodes formed within a semiconductor material proximate to a backside surface of the semiconductor material; a microlens optically aligned with the photosensitive region; a metal shield disposed between the microlens and the photosensitive region, wherein the metal shield partially covers the photosensitive region to define an uncovered portion of the photosensitive region; and a spectral filter disposed between the microlens and the metal shield, wherein the spectral filter partially covers the metal shield to define an uncovered portion of the metal shield.
14. The image sensor of claim 13, wherein a lateral width of the metal shield for each of the plurality of PDAF pixel cells varies based on a position of the individual PDAF pixel cells within the image sensor.
15. The image sensor of claim 13, wherein the PDAF pixel cells include a first PDAF pixel cell associated with a first chief ray angle and a second PDAF pixel cell associated with a second chief ray angle different from the first chief ray angle, wherein the first PDAF pixel cell and the second PDAF pixel cell have a same orientation, and wherein a lateral width of the spectral filter for first PDAF pixel cell is greater than a lateral width of the spectral filter for the second PDAF pixel cell.
16. The image sensor of claim 13, wherein individual imaging pixel cells included in the imaging pixel cells laterally surrounding the individual PDAF pixel cells each include corresponding spectral filters disposed over respective photosensitive regions of the individual imaging pixel cells, and wherein a first photoresponse of the spectral filter included in the individual PDAF pixel cells is different than a second photoresponse of the corresponding spectral filters included in the individual imaging pixel cells.
17. The image sensor of claim 16, wherein individual PDAF pixel cells includes a first PDAF pixel

row or column adjacent to a first imaging pixel cell included in the individual imaging pixel cells, wherein the second spectral filter included in the first imaging pixel cell extends over the uncovered portion of the metal shield included in the first PDAF pixel cell such that the second spectral filter is disposed between the microlens of the first PDAF pixel cell and the metal shield of the first PDAF pixel cell.

18. The image sensor of claim 13, wherein for each of the individual PDAF pixel cells, microlens material included in the microlens fills gaps defined by the uncovered portion of the photosensitive region and the uncovered portion of the metal shield.

19. The image sensor of claim 13, wherein for each of the individual PDAF pixel cells, a first lateral width of the spectral filter is less than a second lateral width of the metal shield along a direction parallel to the backside surface to form a gap extending from an edge of the spectral filter toward an edge of the metal shield when viewed from a plan view.

20. The PDAF pixel cell of claim 19, wherein a third lateral width of the gap along the direction parallel to the backside surface for each of the individual PDAF pixel cells is substantially equal.
